

Dhanush Mathivanan

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Professional Summary

Data Science graduate student at The George Washington University, specializing in Machine Learning, Deep Learning, NLP, Generative AI, and large-scale data systems. AWS Certified cloud engineer with 9+ years of experience building distributed microservices, event-driven architectures, and scalable serverless data platforms on AWS for global enterprise systems, Fullstack web application development. Experienced in developing end-to-end machine learning pipelines using TensorFlow, Scikit-Learn, and FastAPI, with strong expertise in predictive modeling, data analytics, and visualization. Passionate about applying machine learning and cloud-native technologies to solve large-scale real-world data problems.

Technical Skills

Programming: Python, R Studio, SQL, TypeScript, JavaScript, ActionScript

Machine Learning Frameworks: TensorFlow, Scikit-Learn, Hugging Face, Matplotlib, Seaborn

Web Frameworks: Node.js, Express.js, Angular, FastAPI, Streamlit, Terraform, Electron, Redis, NgRx Redux

Databases: DynamoDB, MySQL, MongoDB, SQLite

Cloud & DevOps: AWS (Lambda, CDK, S3, API Gateway, CloudFormation, Cognito, SQS/SNS), Jenkins, Docker

Web Technologies: HTML5, CSS3, JSON, XML, Bootstrap, jQuery

Testing & Debugging: AWS CloudWatch, Dynatrace, Mocha/Chai, Jasmine/Karma, Chrome DevTools

Version Control: Git, SVN

Tools: Tableau Desktop, Workbench, VS-Code, Atom, Pycharm

Education

The George Washington University, Washington, DC

Master of Science in Data Science

Aug 2025 - May 2027

Coursework: Data Analytics, Machine Learning, Deep Learning, NLP, Data Warehousing

Professional Experience

Bosch Global Software Technologies

Client: MHP Porsche (Audi AG), Germany

Technical Project Specialist / Lead Engineer

Mar 2024 - Aug 2025

- Built and deployed Porsche MBP workforce planning platform using AWS serverless microservices architecture with CDKTF, Lambda, API Gateway, DynamoDB, and Cognito, enabling scalable workforce data pipelines that support operational planning for a global automotive enterprise.
- Developed automated event-driven data ingestion pipelines by implementing JSON schema validation (JOI), S3-triggered Lambda processing, and API Gateway integrations, enabling real-time processing of enterprise HR datasets used for large-scale organizational planning decisions.
- Integrated legacy enterprise systems with modern cloud infrastructure through asynchronous event orchestration using SNS, SQS, and Lambda processing pipelines, improving reliability and enabling real-time synchronization of mission-critical workforce data across distributed systems.

Accenture Technologies

Client: Educational Testing Service (ETS), USA

Custom Software Engineering Team Lead

Sep 2021 - Mar 2024

- Reduced candidate exam result processing time by 50% by architecting scalable AWS serverless data pipelines using Lambda, DynamoDB, Cognito authentication, SNS/SQS messaging, and CloudFormation infrastructure templates, improving throughput for a global educational assessment platform processing millions of exam records annually.
- Designed secure event ingestion architecture by validating candidate score events using AJV schema validation and processing events through SQS-triggered Lambda workflows, ensuring reliable real-time processing for high-volume exam scoring systems.

- Improved data platform efficiency and system reliability by developing automated DynamoDB maintenance scripts in Node.js and Python and implementing reusable TypeScript service components, enabling consistent API performance for enterprise-scale educational testing services.

HCL Technologies

Client: Roche Holding AG, Switzerland

Lead Engineer - Full Stack Cloud Development

Nov 2020 - Sep 2021

- Accelerated healthcare diagnostic reporting workflows by 40% by developing RESTful APIs and reporting services using Node.js, Express.js, Angular, and MySQL, enabling faster processing of patient diagnostic data for Roche's global healthcare operations.
- Automated enterprise reporting pipelines by implementing dynamic PDF and Excel report generation using Node.js libraries (ExcelJS, PDF-node) and streaming data pipelines, reducing manual reporting overhead across laboratory testing workflows.
- Enabled secure healthcare system integrations by implementing OAuth/JWT authenticated REST APIs and AWS S3-based storage pipelines, ensuring reliable and compliant exchange of medical workflow data across distributed clinical systems.

Object Frontier Software Pvt Ltd

Client: International Flavours Fragrance (IFF), USA

Software Engineer - Full Stack Development

Aug 2018 - Nov 2020

- Developed offline-first enterprise desktop application using Electron framework with embedded Node.js server and SQLite/MySQL synchronization architecture, enabling uninterrupted operations for global manufacturing and supply chain workflows.
- Improved application scalability and user transaction management by implementing Angular ngrx-redux state architecture with reducer-action data flow patterns, enabling advanced features such as undo/redo operations across complex enterprise transactions.
- Strengthened backend reliability and database performance by integrating Node.js microservices with MySQL through Sequelize ORM and implementing structured logging using Winston, improving system observability and production stability for enterprise-grade systems.

V2Soft Pvt Ltd

Client: Fiat Chrysler Automobiles (FCA), USA

Software Engineer - HMI S/W Development

Oct 2016 - Aug 2018

- Developed responsive automotive Human-Machine Interface dashboards using Angular, HTML5, Bootstrap, and modular JavaScript components, improving usability across vehicle infotainment and diagnostic systems.
- Enabled scalable vehicle telemetry data processing by implementing MongoDB batch jobs and Node.js backend APIs, supporting ingestion and analysis of automotive diagnostic logs across vehicle platforms.
- Implemented real-time event communication by developing Socket.IO-based messaging between backend Node.js services and frontend dashboard modules, enabling real-time monitoring of vehicle operational data.

Certifications & Awards

Phi Delta Gamma Fellowship recipient - George Washington University (2026)

Global Leaders Fellowship recipient - George Washington University (2025)

Introduction to R (DataCamp) (2025)

AWS Certified Solutions Architect, Associate - SAA-C03 (2024)

AWS Certified Developer, Associate - DVA-C02 (2023)

Client Value Creation Award, Best People Award - Accenture (2022)

Data Analysis Projects

- Heart Disease prediction based on US CDC Survey Data (2024): Code Base Github , ML Trained Predictor Web App
- Health Status prediction based on Income factors in US Population CDC Survey Data (2024): Code Base Github